# American Computer Science League

2025-2026 ● Contest 3: Short Problems ● Intermediate Division

**1. Boolean Algebra**

Simplify the following Boolean algebra expression:

$$A(\overline{B + C}) + \overline{\overline{BC}} + C$$

A. $B + C$
B. $\overline{B} + C$
C. $B + \overline{C}$
D. $\overline{B} + \overline{C}$
E. $B\overline{C}$

**2. Boolean Algebra**

How many ordered triples make the following Boolean algebra expression TRUE?

$$\overline{A(\overline{B + C}) + \overline{B + \overline{C}}}$$

A. 3
B. 5
C. 6
D. 7
E. 8

**3. Data Structures**

Build a binary search tree for the string (Left subtree $\leq$ root $<$right subtree)

CALIFORNIA

What is the internal path length?

A. 17
B. 19
C. 20
D. 21
E. 23

**4. Data Structures**

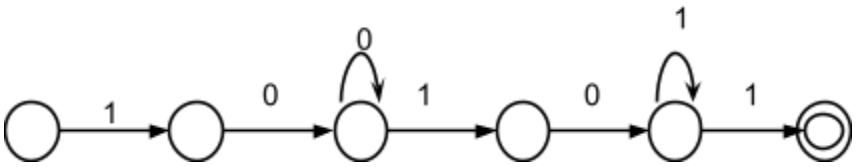
Given an initially empty queue and the following commands on the queue, what item will be popped next?

```
PUSH("S")
PUSH("A")
PUSH("C")
x = POP()
PUSH("R")
PUSH("A")
x = POP()
x = POP()
PUSH("M")
PUSH("E")
x = POP()
PUSH("N")
PUSH("T")
PUSH("O")
x = POP()
```

A. N
B. T
C. O
D. E
E. M

**5. FSAs and Regular Expressions**

What is the regular expression for the following FSA below?




A. 1001011
B. (10101)*
C. 10*101*
D. 10*0011*1
E. 100*101*1

**6. FSAs and Regular Expressions**

How many of the following words are accepted by this regular expression?

```
[a-ct-v][e-m|o-u]*[aeiou]
```

| | | |
|---|---|---|
| violin | cello | trombone |
| trumpet | oboe | saxophone |
| tuba | flute | clarinet |

A. 1
B. 2
C. 3
D. 4
E. 5

# American Computer Science League

2025-2026 ● Contest 3: Short Problems ● Intermediate Division

**1. 布尔代数**

请简化**下列布尔表达式**

$$A(\overline{B + C}) + \overline{\overline{BC}} + C$$

A. $B + C$
B. $\overline{B} + C$
C. $B + \overline{C}$
D. $\overline{B} + \overline{C}$
E. $B\overline{C}$

**2. 布尔代数**

**请问有多少个有序三元组可以使得下列表达式的结果为真？**

$$\overline{A(\overline{B + C}) + \overline{B + \overline{C}}}$$

A. 3
B. 5
C. 6
D. 7
E. 8

**3. 数据结构**

**为以下字符串构建一棵二叉搜索树。** （左子树≤根、右子树 > 根）

CALIFORNIA

**请问**其内部路径长度是多少？

A. 17
B. 19
C. 20
D. 21
E. 23

**4. 数据结构**

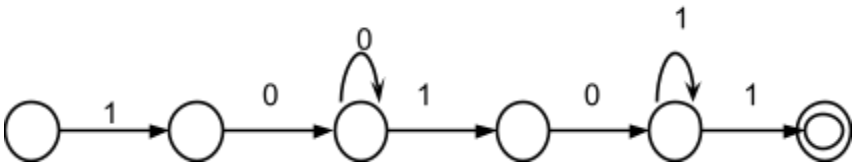
给定一个初始为空的队列，以及对队列执行以下操作指令后，请问下一个被弹出的元素是什么？

```
PUSH("S")
PUSH("A")
PUSH("C")
x = POP()
PUSH("R")
PUSH("A")
x = POP()
x = POP()
PUSH("M")
PUSH("E")
x = POP()
PUSH("N")
PUSH("T")
PUSH("O")
x = POP()
```

A. N
B. T
C. O
D. E
E. M

**5. FSAs 和 正则表达式**

请问以下FSA的正则表达式是什么？




A. 1001011
B. (10101)*
C. 10*101*
D. 10*0011*1
E. 100*101*1

**6. FSAs 和 正则表达式**

请问下列字符串中有多少个能够被该正则表达式接受？

```
[a-ct-v][e-m|o-u]*[aeiou]
```

| | | |
|---|---|---|
| violin | cello | trombone |
| trumpet | oboe | saxophone |
| tuba | flute | clarinet |

A. 1
B. 2
C. 3
D. 4
E. 5